

LESSON 1 · 00:00

Regression is a model finder

A regression chooses parameter values that make an equation match data. In Desmos, use a tilde, not an equals sign. The subscripts x_1 and y_1 refer to table columns.

$$y_1 \sim mx_1 + b$$

f

CHECK

Today: choose, fit, interpret, verify

01 • TABLE BASICS

Build the table correctly

Enter x-values in the first column and y-values in the second. Keep units consistent. One mistyped row can change every fitted parameter.

x_1	y_1
0	5
2	11
4	17
6	23
8	29

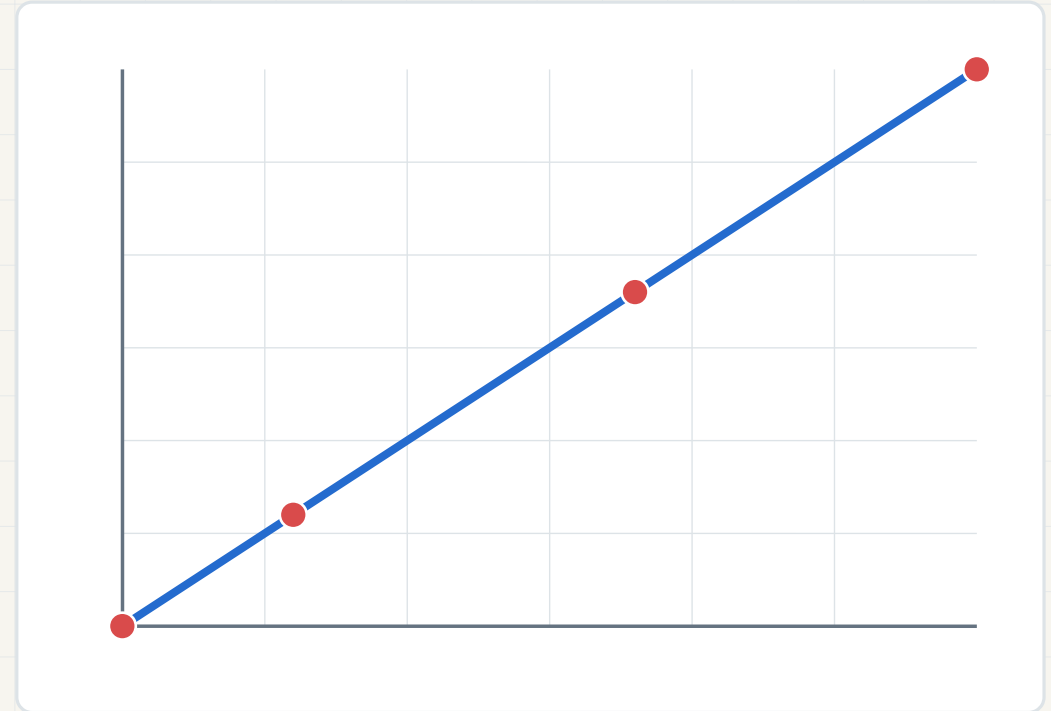
CHECK

Pattern: +3 in y for every +1 in x

GUIDED EXAMPLE

Exercise 1 • Linear model

A bike rental costs the amounts shown. Use a linear regression to find the hourly rate and initial fee.



CHECK

$$m = 4.5 \text{ and } b = 8$$

MEANING > BUTTONS

Interpret the parameters

The slope is the change in cost per hour.
The intercept is the cost at zero hours.
Always answer in the units of the story.

$$C = 4.5h + 8$$

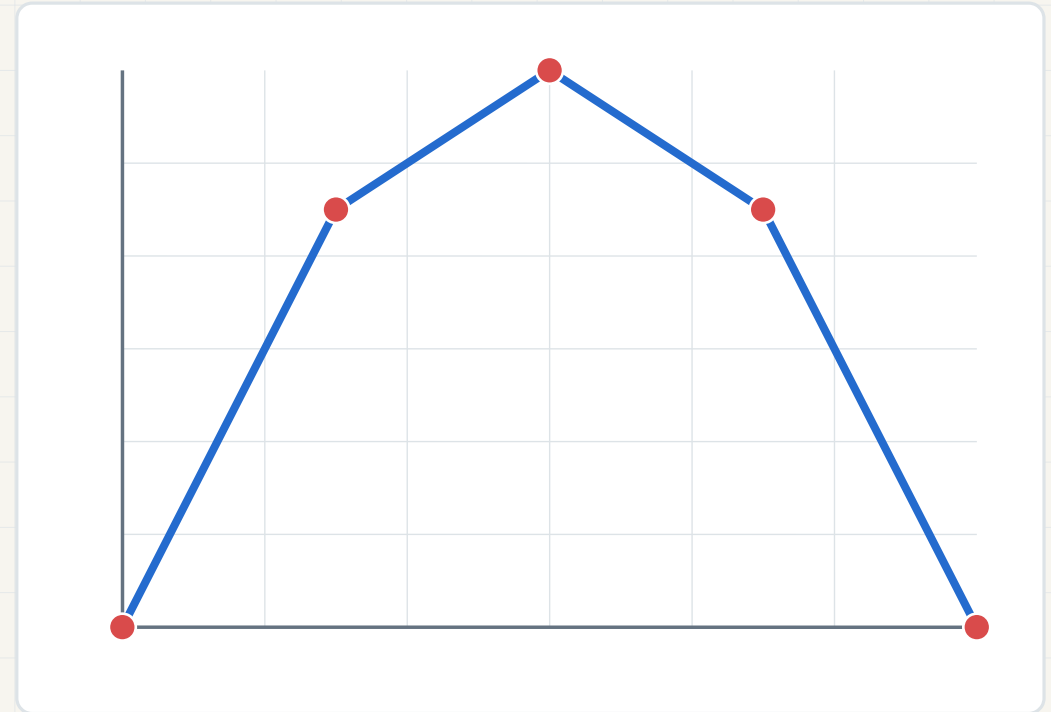
f

CHECK **\$4.50 per hour; \$8 initial fee**

TRY IT • PAUSE 40 SECONDS

Exercise 2 • Quadratic model

A ball is launched. The table gives height y , in feet, after x seconds. Fit a quadratic and find the maximum height.



CHECK

 $a = -16$, $b = 64$, $c = 5$; maximum = 69 ft

SOLUTION

Vertex from the fitted equation

For $y = ax^2 + bx + c$, the vertex occurs at $x = -b/(2a)$. Substitute the fitted values, then evaluate the function. Desmos also shows the vertex when you click the curve.

$$x = -64 \div (2 \cdot -16) = 2$$
$$y(2) = 69$$

f

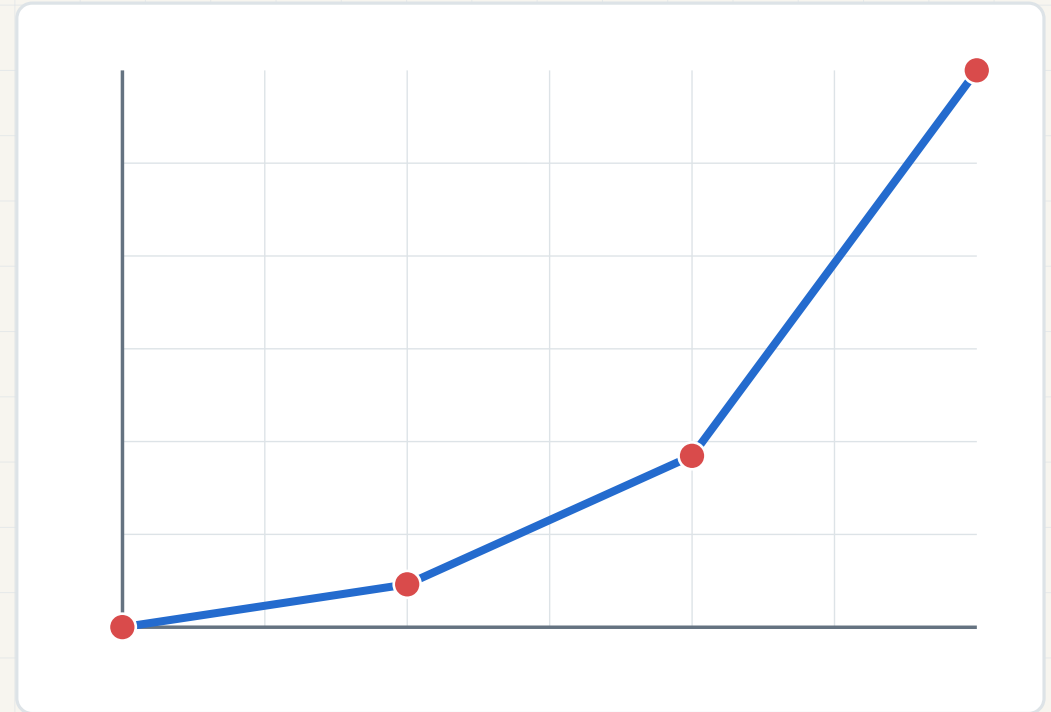
CHECK

The ball peaks at 69 feet after 2 seconds

MODEL CHOICE

Exercise 3 · Exponential growth

A culture starts with 120 cells and triples every 2 hours. Fit $y = ab^x$. Determine the value after 5 hours.



CHECK

$$a = 120, b = \sqrt{3} \approx 1.732; y(5) \approx 1,871$$

DECISION RULE

Linear or exponential?

Use linear when equal x-steps produce equal differences. Use exponential when equal x-steps produce equal ratios. A curved scatterplot alone is not enough evidence.

Differences → linear

Ratios → exponential

CHECK

Check the table before choosing the model

INDEPENDENT PRACTICE · PAUSE 45 SECONDS

Exercise 4 · Prediction

A town records population in thousands: 18.0, 19.4, 20.9, 22.5 at years 0, 1, 2, 3. Fit an exponential model and predict year 6.

x_1	y_1
0	18
1	19.4
2	20.9
3	22.5

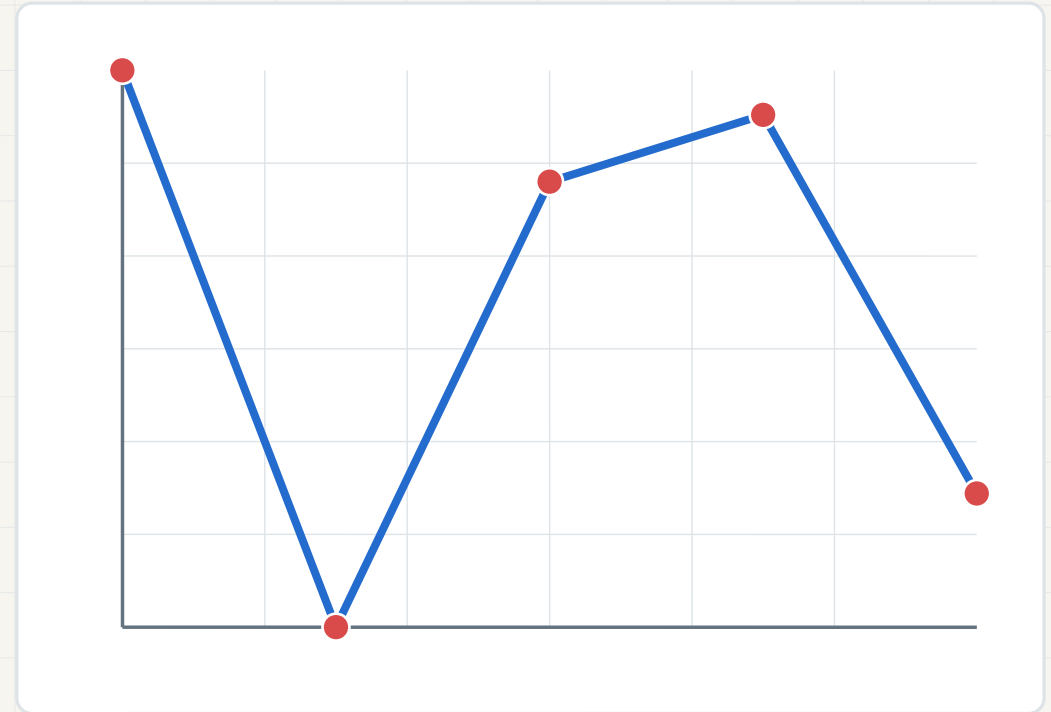
CHECK

 $a \approx 17.99$, $b \approx 1.077$; year 6 ≈ 28.1 thousand

DON'T SKIP THE CHECK

Residuals reveal fit quality

A residual equals actual minus predicted. Random small residuals support the model. A curved residual pattern warns that the model type is wrong.

**CHECK****Near zero is good; pattern-free is better**

SAT-STYLE CHALLENGE

Mixed checkpoint

The model $P = 240(1.06)^t$ predicts a balance after t years. What does 1.06 mean, and when does P first exceed 300?

$$240(1.06)^t > 300$$

f

CHECK

6% yearly growth; first exceeds \$300 during year 4

YOUR THREE-STEP ROUTINE

Lesson 1 recap

One: enter the table accurately. Two: choose a model from differences, ratios, or context. Three: interpret parameters and verify the prediction in context.

**TABLE → MODEL →
MEANING**

f

CHECK

Next: use regressions to solve unknown constants